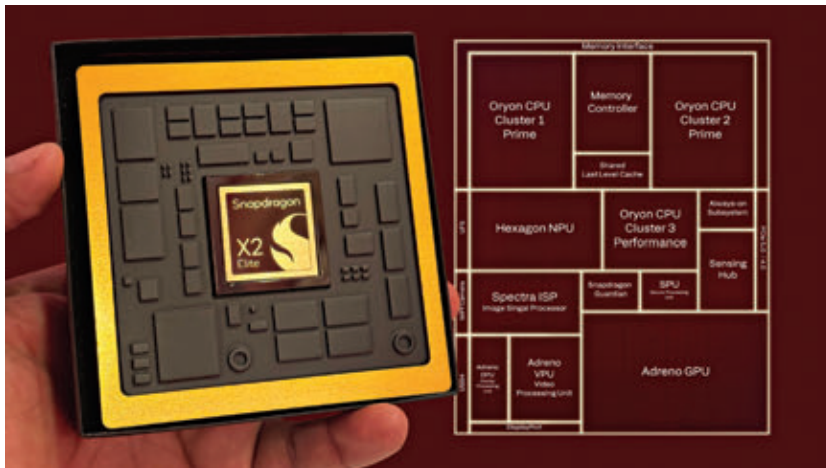




Designing a world-class CPU

Qualcomm's VP of CPU Engineering, **Pradeep Kanapathipillai** on a new chapter in PC-class compute

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When I ask what has changed between the first-generation Oryon CPU and the latest one inside X2 Elite, Pradeep doesn't hesitate. "Yes, dramatically changed," he says. "Over 100+ architecture and feature enhancements just in the CPU subsystem alone," covering everything from IPC and ISA improvements to power-efficiency optimizations, plus the addition of SME – "the matrix accelerator I spoke about."

In Pradeep's view, the RISC vs CISC debate has largely faded. The ISA differences, he says, "pale in comparison to everything else we need." Real competition now lies in orchestrating memory paths, prediction engines, security layers, and power states – the deep plumbing of modern CPUs.

From mobile to AI PC

If Oryon's DNA feels shaped by Qualcomm's mobile heritage, that's no

accident. Moving from mobile SoCs to laptop-class compute sounds straightforward, but Pradeep calls it a multi-year leap. "Learnings had to do with energy efficient computing, low power design," he says, but warns that "there isn't a silver bullet... it all boils down to hard engineering choices," distributed across "over a 1000 features."

TDP is where the transition truly changes the rules. Phones operate around 8 watts; PCs begin "from 10-watt onwards and upwards to many tens of watts." To survive there, a CPU must expand in both directions. Supporting "differing TDPs... for different product SKUs" requires a suite of power-management systems manufacturers can tune at launch. He points to "limit management" – setting power, thermal, and current ceilings – and adds, "There are at least 25 such major architectures that have to be built." Part of that discipline involves physical

placement – the dark art of SoC floor-planning. "Latency to DRAM and all the intermittent cache... matters greatly," he says, with prime cores placed closest to DRAM through the most direct paths.

Different from Intel, Apple

One architectural choice that stands out is the absence of efficiency cores. "The power performance profile for the Performance cores is higher than that of some of the E-cores you might be referring to," Pradeep says. For laptops, "It makes sense to have higher performance cores than true low performance energy efficient cores."

But the heart of Oryon's story is that it was built from scratch. "We start with a clean sheet of paper and the goal is to bring out a market leading CPU," he says, one that must compete with "Intel and AMD who have been in this business for 20+ years." Clean-sheet CPU design allows Qualcomm to "microarchitect to ensure there aren't any side-channel vulnerabilities." After five years, he sums it up simply: "It's been net positive."

Security is one area where that freedom shows. Qualcomm has "strengthened" side-channel mitigations, reinforced MMU protections, and "strengthened" several cryptographic ciphers. Much of this work is invisible, but essential.

Oryon also isn't confined to PCs. "We share a lot of common technologies across business units, but customize it for various markets," he says. Oryon is a framework – architectural DNA that adapts to phones, XR, automotive, and industrial systems.

Advice for young engineers

When asked what he'd tell India's young engineers aspiring to join semiconductors, he's direct. "You have to be an expert in at least one subsystem such as CPU, NPU, GPU," while building breadth across SoC components. VLSI, he says, "will be in demand for a long time to come." **d**